

Harshit Singh

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SUMMARY

Computer Science undergraduate at IIT Jammu with hands-on cybersecurity experience across security monitoring, vulnerability assessment, network analysis, and log investigation. Trained in SOC Level 1 workflows (threat detection, IDS/Snort, traffic and log analysis with Wireshark) and comfortable across Windows, Linux, TCP/IP, and DNS fundamentals. Eager to support a financial-services Information Security team in monitoring, vulnerability management, risk assessment, and compliance.

EDUCATION

B.Tech — Computer Science & Engineering, IIT Jammu

2024 – 2028

Relevant Coursework: Computer Networks, Operating Systems, Cybersecurity Foundations, Data Structures & Algorithms, Computer Architecture.

TECHNICAL SKILLS

Security Monitoring	Snort (IDS / traffic logging), Wireshark, log & event analysis, threat detection
Vulnerability Mgmt.	Vulnerability assessment, CVE research, Exploit-DB, Nmap, Gobuster, remediation tracking
Networking	TCP/IP, DNS, OSI Model, network security fundamentals, packet analysis
Operating Systems	Linux (admin, CLI, scripting), Windows
Tools	Burp Suite, Metasploit (basic), LinPEAS, Docker
Programming	Python, Bash, C/C++
Concepts	Risk assessment, security documentation & policy, IAM basics, cryptography (AES, RSA), OSINT, forensics

EXPERIENCE

Research Intern — Network Security & Measurement | IIT Bombay

May 2026 – Present

- Analyze large-scale network traffic to detect anomalies, filtering behavior, and security-relevant patterns in real-time communication protocols.
- Build analysis pipelines to process and correlate large network datasets, identifying traffic anomalies, threat indicators, and filtering behavior (Tracebox, Quack/HyperQuack); document findings for review.

Coding Club Department Head | IIT Jammu

Aug 2025 – May 2026

- Delivered technical workshops on network analysis, vulnerability assessment, and security best practices to 50+ students.
- Designed and administered Capture-The-Flag competitions, including challenge infrastructure, secure hosting, and scoring systems.
- Authored challenge documentation and audited environments for unintended exposure — mirroring real security review and remediation work.

PROJECTS & RESEARCH

CTF Infrastructure & Challenge Design (Nexus 2026, Anhad 2026) | IIT Jammu

Jan 2026

- Built Web, OSINT, and Forensics challenges simulating real attack surfaces (XSS, SQLi, IDOR) to demonstrate common application vulnerabilities.
- Deployed secure, containerized hosting with Docker; audited for misconfigurations, flag leakage, and unintended solution paths.
- Produced setup and remediation documentation for each challenge environment.

TryHackMe Command Validator + VM | Personal Project

May 2026 – Present

- Built a learning platform with real-time shell-session tracking and a Python/Bash validation engine that verifies task completion against system state.
- Reinforced practical skills in endpoint monitoring, log inspection, and Linux administration.

UAV Security Assessment Framework | Ongoing Research

Dec 2025 – May 2026

- Developing a vulnerability-assessment framework combining scripted CVE analysis with communication-layer testing; conducting structured attack simulations and documenting risk findings.

CERTIFICATIONS & TRAINING

- TryHackMe Paths:** Pre Security, Web Fundamentals, Cyber Security 101, SOC Level 1 (in progress).
- Key Modules:** Snort (IDS), Wireshark, Defensive Security, Cyber Threat Intelligence, Networking, HTTP/HTTPS analysis, Cryptography, Linux Fundamentals, Intro to Pentesting.
- Cybersecurity Workshop — Yhills India.

CTF PARTICIPATION & ACHIEVEMENTS

- Competed in BlitzCTF 2025, HOLMES CTF 2025, Razz Security 2025, University CTF 2025, and Industrial Intrusion CTF 2025 (web, forensics, crypto, OSINT); designed and administered live CTF machines, demonstrating exploits and defenses in competition environments.